BUSINESS REPORT

FINANCE AND RISK ANALYTICS
PART-B

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Problem

PART A

Problem Statement:

The dataset contains 6 years of information (weekly stock information) on the stock prices of 10 different Indian Stocks. Calculate the mean and standard deviation on the stock returns and share insights. You are expected to do the Market Risk Analysis using Python.

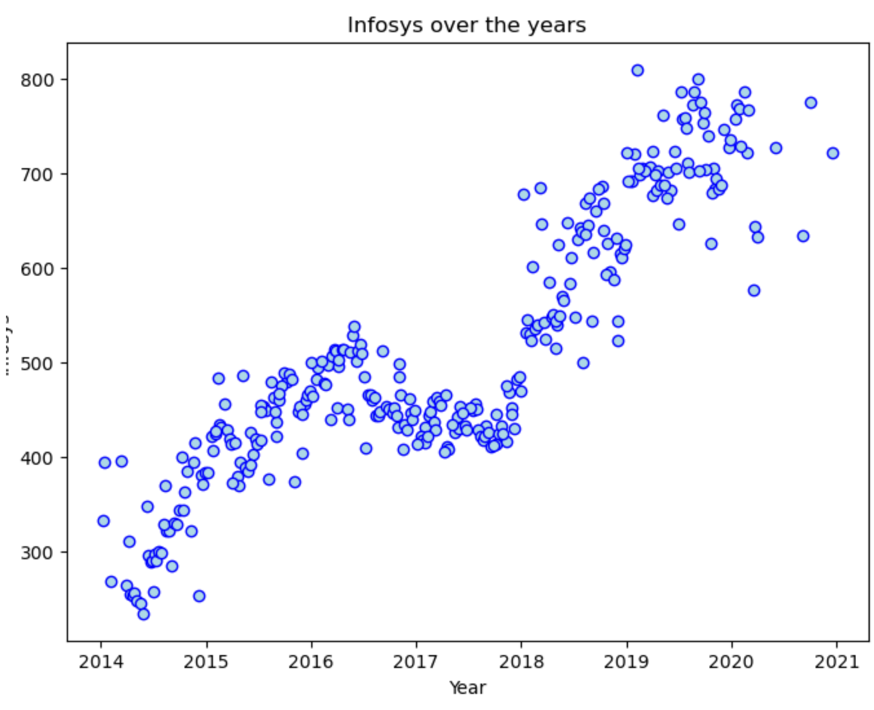
Dataset for problem: Market Risk Dataset

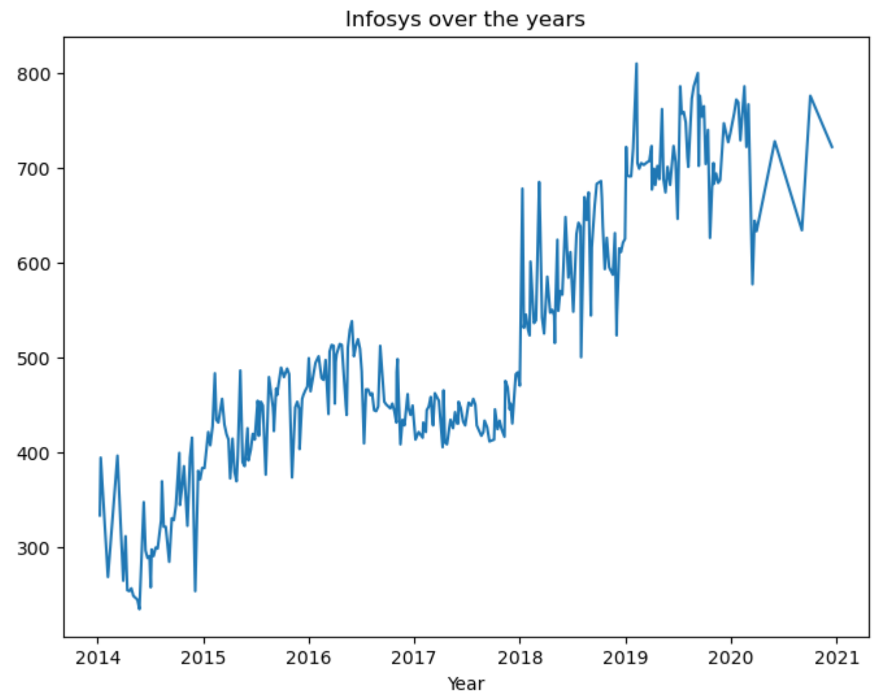
1.1 Draw Stock Price Graph (Stock Price vs Time) for any 2 given stocks with inference:

The data having 314 rows and 11 columns has been read. The data pertains to Stock Prices of 10 Stocks for different dates ranging from 31.03.2014 to 30.03.2020. The data was checked for Null values and Duplicates and it was found that there were no Null values or Duplicates in the data.

Stock Price Graphs of two Stocks viz. Infosys and Vodafone Idea have been drawn and the inferences are as under:

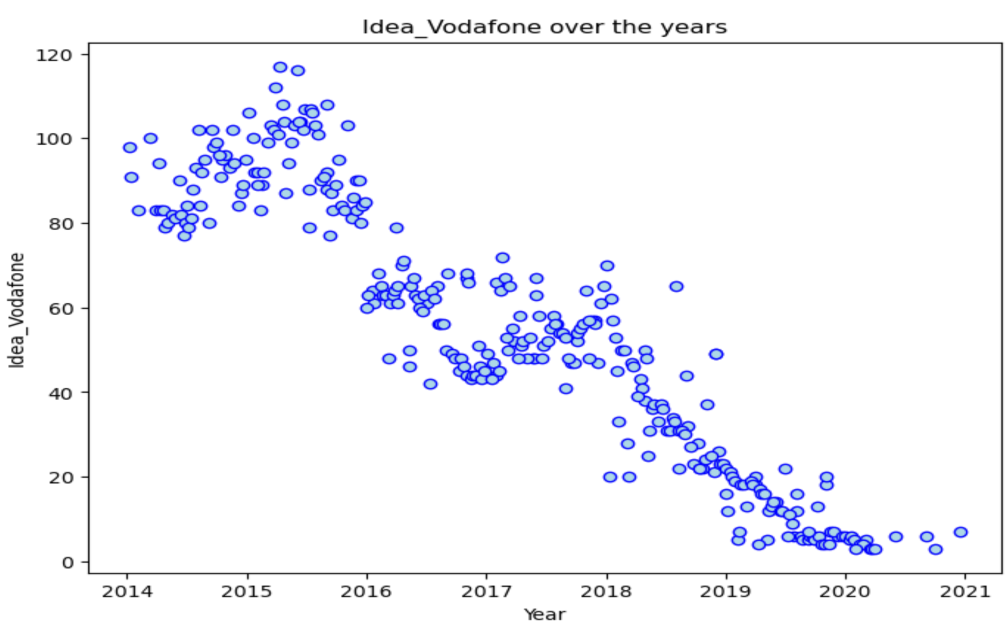
<u>Scatter Plot and Line Chart of Infosys:</u>

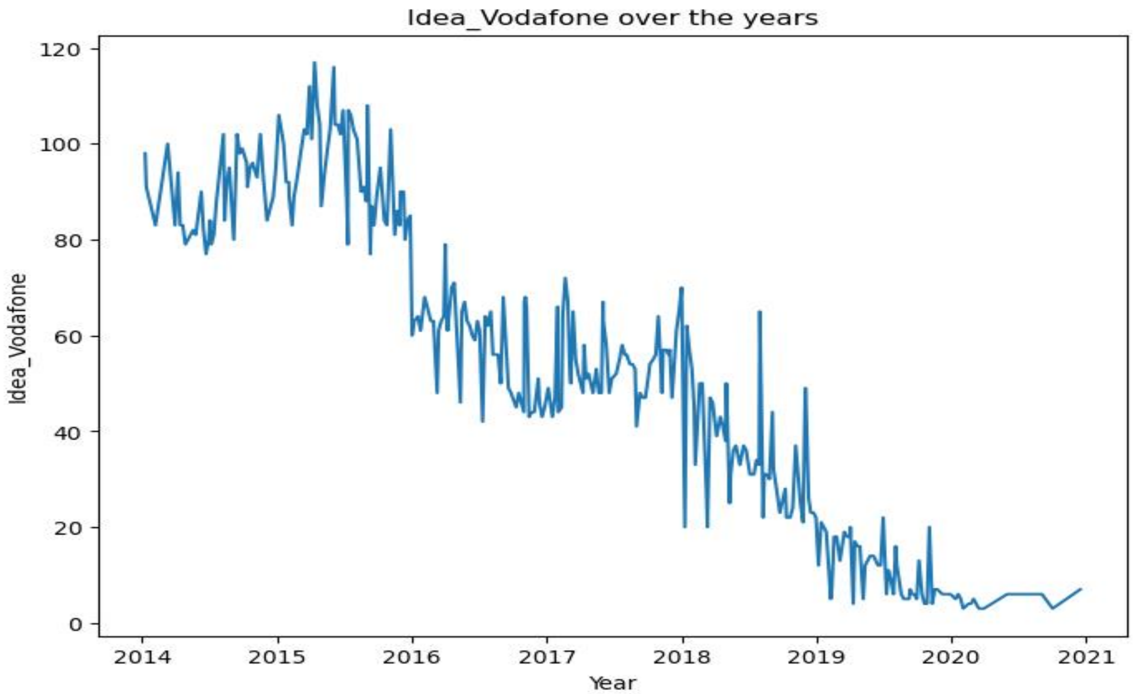




Inference about Infosys Stock: The stock of Infosys is steadily rising since 2014 through 2021. The stock had a no growth period during 2017-18, which can be explained due to US stock market crash in 2008 leading to global financial crisis. The stock of Infosys further had a no growth period and a dip in 2020 and 2021, which can be explained due to corona pandemic.

Scatter Plot and Line Chart of Idea Vodafone:





Inference about Idea Vodafone Stock: The stock of Idea Vodafone remained steady from 2014 to 2016. After 2016, Idea Vodafone always had governance issues and remained cash strapped. It failed to make timely investment in the network and rollout of 4G services in the country. Due to above reasons, Vodafone Idea has been steadily losing customers leading to losses and this is the reason of its continuous fall since 2016.

1.2 Calculate Returns for all stocks with inference:

The returns of all the 314 stocks have been calculated using Python and the results in the form of Head and Tail of Data are as under:

```
stock_returns.head()
```

	Infosys	Indian_Hotel	M&M	Axis_Bank	SAIL	Shree_Cement	Sun_Pharma	Jindal_Steel	Idea_Vodafone	Jet_Airways
0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	-0.026873	-0.014599	0.006572	0.048247	0.028988	0.032831	0.094491	-0.065882	0.011976	0.086112
2	-0.011742	0.000000	-0.008772	-0.021979	-0.028988	-0.013888	-0.004930	0.000000	-0.011976	-0.078943
3	-0.003945	0.000000	0.072218	0.047025	0.000000	0.007583	-0.004955	-0.018084	0.000000	0.007117
4	0.011788	-0.045120	-0.012371	-0.003540	-0.076373	-0.019515	0.011523	-0.140857	-0.049393	-0.148846

```
stock_returns.tail()
```

	Infosys	Indian_Hotel	M&M	Axis_Bank	SAIL	Shree_Cement	Sun_Pharma	Jindal_Steel	Idea_Vodafone	Jet_Airways
309	0.009649	-0.110348	0.030305	-0.057580	-0.087011	0.023688	0.072383	-0.053346	-0.287682	-0.127833
310	-0.139625	-0.051293	-0.093819	-0.145324	-0.095310	-0.081183	-0.043319	-0.187816	0.693147	-0.200671
311	-0.094207	-0.236389	-0.285343	-0.284757	-0.105361	-0.119709	-0.050745	-0.141830	-0.693147	-0.117783
312	0.109856	-0.182322	-0.091269	-0.173019	-0.251314	-0.067732	-0.076851	-0.165324	0.000000	-0.133531
313	-0.017228	0.000000	-0.031198	0.051432	0.090972	-0.006816	0.040585	-0.081917	0.000000	0.000000

Inference**:** All 10 stocks have positive and negative returns in different cycles. This explains the cyclic nature of the stock market. All growth stocks such as Shree Cement, Infosys, Axis Bank and Indian Hotels etc. have more positive cycles whereas lagging stocks such as Jet Airways, Idea Vodafone and Jindal Steel etc. have more number of negative cycles during the period.

1.3 Calculate Stock Means and Standard Deviation for all stocks with inference:

The Mean Returns and Standard Deviation of all the Stocks have been calculated using Python and results are as under:

Mean Returns of all Stocks:

```
Infosys          0.002794
Indian_Hotel     0.000266
M&M             -0.001506
Axis_Bank        0.001167
SAIL            -0.003463
Shree_Cement     0.003681
Sun_Pharma      -0.001455
Jindal_Steel    -0.004123
Idea_Vodafone   -0.010608
Jet_Airways     -0.009548
dtype: float64
```

Standard Deviation of all Stocks:

```
Infosys          0.035070
Indian_Hotel     0.047131
M&M              0.040169
Axis_Bank        0.045828
SAIL             0.062188
Shree_Cement     0.039917
Sun_Pharma       0.045033
Jindal_Steel     0.075108
Idea_Vodafone    0.104315
Jet_Airways      0.097972
dtype: float64
```

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Average Returns vs Volatility of all Stocks:

	Average	Volatility
Infosys	0.002794	0.035070
Indian_Hotel	0.000266	0.047131
M&M	-0.001506	0.040169
Axis_Bank	0.001167	0.045828
SAIL	-0.003463	0.062188
Shree_Cement	0.003681	0.039917
Sun_Pharma	-0.001455	0.045033
Jindal_Steel	-0.004123	0.075108
Idea_Vodafone	-0.010608	0.104315
Jet_Airways	-0.009548	0.097972

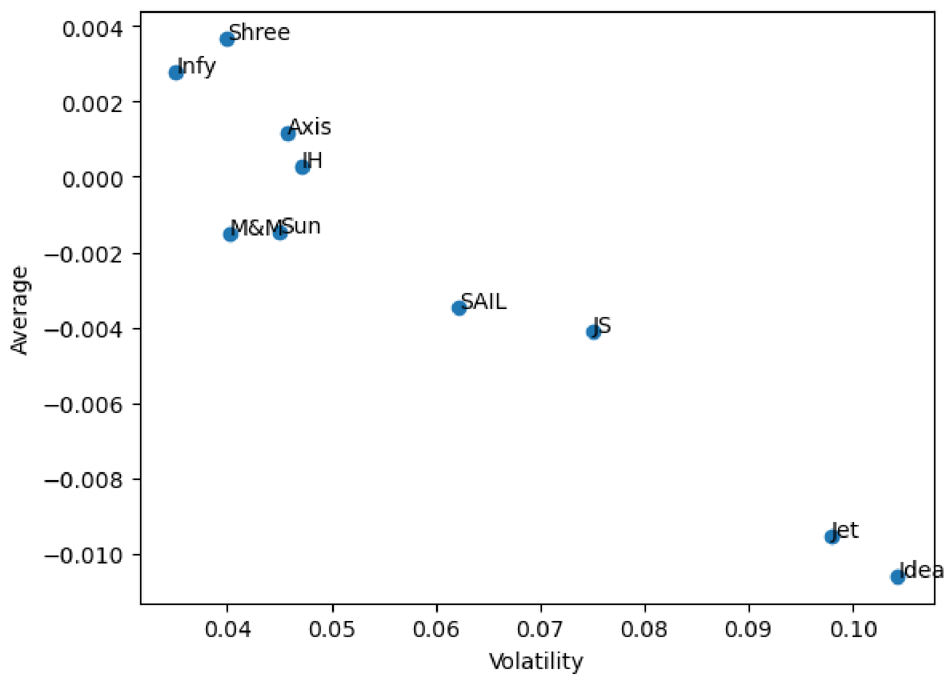
Inference:

i) Average Returns (Mean) of Shree Cement is the highest during the period and that of Idea Vodafone is the lowest during the period.

ii) Volatility (Standard Deviation) of Idea Vodafone is highest during the period and that of Infosys is lowest during the period.

1.4 Draw a plot of Stock Means vs Standard Deviation and state your inference:

Stock Means (Average) and Standard Deviation (Volatility) have been plotted using Python (Scatter Plot) and the plot is as under:



Inference: Shree Cement and Infosys are the two stocks having maximum average returns and minimum volatility. Whereas, Vodafone Idea and Jet Airways are the two stocks having minimum average returns and maximum volatility.

1.5 Conclusions and Recommendations:

The Stocks are evaluated on the basis of their Mean and Standard Deviation. Where, Mean signifies average returns of a period of the Stock. Higher positive average returns means that the stock has given positive returns over the time whereas negative average returns means that there has been negative returns (loss) during the period. In the given data, average returns of Shree Cement, Infosys, Axis Bank and Indian Hotels are positive whereas average returns of Idea Vodafone, Jet Airways, Jindal Steel, Sun Pharma, SAIL and M&M are negative during the period.

Standard Deviation helps determine market volatility (risk) of a stock. Stocks having high Standard Deviation are highly volatile, whereas stocks with low Standard Deviation are steady and come with low risk. In the given data, Infosys and Shree Cement are having least Standard Deviation, therefore they are least volatile (having low risk) and Stocks of Idea Vodafone and Jet Airways are having high Standard Deviation, therefore they are highly volatile (having high risk).